120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 10: Advanced Machine Learning Concepts

Day 65: Bias–Variance Tradeoff

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1 Introduction

One of the most important concepts in machine learning is the **bias–variance tradeoff**. It explains the fundamental challenge in building predictive models: balancing model complexity with the ability to generalize to unseen data.

In financial prediction tasks, this concept is especially important because market data is noisy and models can easily overfit historical data.

2 Prediction Setup

Assume that the true relationship between features and the target variable is:

$$y = f(x) + \epsilon$$

where:

- $f(x)$ is the true underlying function
- ϵ is random noise
- $\mathbb{E}[\epsilon] = 0$
- $Var(\epsilon) = \sigma^2$

We attempt to estimate $f(x)$ using a model:

$$\hat{f}(x)$$

The quality of the model can be measured using the expected prediction error.

3 Expected Prediction Error

For a given input x, the expected squared prediction error is:

$$E[(y - \hat{f}(x))^2]$$

Substituting the data-generating process:

$$y = f(x) + \epsilon$$

we obtain:

$$E[(f(x) + \epsilon - \hat{f}(x))^2]$$

This expression can be decomposed into three components.

4 Bias–Variance Decomposition

The expected prediction error can be written as:

$$E[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

Where:

- Bias measures how far the average model prediction is from the true function.
- Variance measures how much the model prediction changes across different training datasets.
- σ^2 is the irreducible noise in the data.

5 Bias

Bias measures the systematic error introduced by approximating the true function.

$$Bias(x) = E[\hat{f}(x)] - f(x)$$

High bias means the model is too simple and fails to capture the underlying pattern.

Examples of high bias models include:

- Linear models applied to nonlinear data
- Overly simplified regression models

High bias typically leads to **underfitting**.

6 Variance

Variance measures how sensitive the model is to fluctuations in the training data.

$$Var(x) = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

High variance means the model fits the training data very closely but may not generalize well.

Examples of high variance models include:

- Deep decision trees
- Highly flexible models

High variance typically leads to **overfitting**.

7 Bias–Variance Tradeoff

Increasing model complexity generally leads to:

- Decreasing bias
- Increasing variance

Conversely, simpler models have:

- Higher bias
- Lower variance

The goal is to find a balance that minimizes the total prediction error.

8 Graphical Intuition

A typical bias–variance curve shows:

- Training error decreasing with model complexity
- Test error first decreasing then increasing

The optimal model complexity occurs at the minimum test error.

9 Example in Financial Modeling

Consider predicting stock returns using market features:

$$r_{t+1} = f(X_t) + \epsilon_t$$

Possible models include:

- Linear regression (low variance, higher bias)
- Decision trees (low bias, high variance)
- Random forests (balanced bias and variance)

Financial data contains significant noise, which makes controlling variance particularly important.

Regularization techniques such as Ridge and Lasso regression help reduce variance by penalizing large coefficients.

10 Role of Model Complexity

Model complexity determines the bias–variance balance.

Examples:

Model	Bias	Variance
Linear Regression	High	Low
Decision Trees	Low	High
Random Forests	Medium	Medium

Ensemble methods such as random forests and boosting are often used to improve this tradeoff.

11 Implications for Quantitative Finance

In financial modeling:

- Data is noisy
- Signal-to-noise ratio is low
- Overfitting is a major risk

Therefore controlling variance through regularization, cross-validation, and ensemble models is essential.

Understanding the bias–variance tradeoff helps researchers design models that generalize better to unseen market conditions.

12 Conclusion

The bias–variance tradeoff highlights a fundamental limitation in machine learning:

- Simple models may miss important patterns (high bias)
- Complex models may overfit the data (high variance)
- The best model balances these two sources of error

In quantitative finance, achieving this balance is critical for building robust trading strategies and predictive models.